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Special Collection: Atomic Layer Deposition (ALD)

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ABSTRACT

Metal phosphates are gaining attention as versatile functional materials with a wide range of applications. Depositing these materials as thin conformal layers with atomic layer deposition requires a phosphorous precursor. Here, the use of tris(dimethylamino)phosphine (TDMAP) is explored by reporting the characterization of an aluminum phosphate process using TDMAP, the well-known aluminum precursor trimethylaluminum (TMA), and O_2-plasma as precursors. Films grown with a four-step process (TDMAP–O_2-plasma–TMA–O_2-plasma) at 300 °C have a stoichiometry of $Al_1P_{0.51}O_{2.9}$ and are amorphous, but do contain $[PO_4]^{3-}$ units. Comparing the four-step process with three-step processes omitting one of the two O_2-plasma steps provides insights into the reactivity of precursors. TMA is able to react with a TDMAP terminated surface but not vice versa. TDMAP has low inherent reactivity and requires the use of a coreactant to remove the ligands of the adsorbed TMA molecules and render the surface reactive for TDMAP adsorption. By implementing a supercycle approach using the (TDMAP–O_2-plasma) sequence and the (TMA–O_2-plasma) sequence as subcycles, more phosphorous rich films can be obtained. The phosphorous content of the deposited aluminum phosphate is shown to have a significant effect on its performance as a battery coating material.

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I. INTRODUCTION

With increasing complexity of materials and processes across various manufacturing industries, the need to have specialized techniques for creating these materials and tailoring their production processes has also increased. Atomic layer deposition (ALD) is one of the most interesting techniques for the deposition of thin films. The characteristics of the final film can be highly customized to suit the requirements of the desired final implementation. This is realized by offering control over a variety of film properties such as composition, crystalline phase, thickness, and conformality. These customization options play a vital role in applications such as, among others, catalysis,[1–3] battery electrode coating materials,[4] and solid electrolytes.[5]

One class of materials within the field of ALD that is currently gaining interest are metal phosphates.[6] These have well established uses within energy storage devices[7] as well as more recent applications in the field of (electro)catalysis.[8] Applying a thin metal phosphate coating on electrodes has been shown to increase the cyclability of Li-ion batteries by reducing undesired reactions between the electrolyte and electrode surface.[9,10]

These favorable conditions can be attributed to some of the inherent properties of phosphates. The phosphate polyanion ($[PO_4]^{3-}$) is more stable than its oxide counterpart (O^{2-}); it can spread its charge over multiple resonance structures, which delocalizes the electron density and, thus, provides more stability. When

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used as an electrode material for a Li-ion battery, they provide a higher potential than oxides (of the same transition metal) due to the inductive effect of the polyanion.[11] They also provide intrinsic proton conduction[12] due to their acidic groups ($H_2PO_4^-$, HPO_4^{2-}) and by shuttling protons through free -OH groups attached to the phosphorous center.[13]

Currently, the phosphorous precursor that is most widely used is trimethyl phosphate (TMP). This is a volatile, stable, and relatively cheap precursor, which has been successfully used to deposit a wide variety of metal phosphate materials by ALD.[6,14,15] TMP, however, is inherently not very reactive toward organometallic compounds and, therefore, three- or four-step ALD processes are typically used in which an oxidizing coreactant is pulsed before and/or after the TMP exposure to remove the nonreactive ligands of the adsorbed organometallic or TMP precursor. The resulting films have a limited phosphorous content (e.g., P/M of 0.7; O/P of 4.3 for aluminum phosphate), and a growth per cycle of 0.18 nm/cycle was reported for aluminum phosphate deposited at 300 °C by Hornsveld *et al.*[14]

Dobbelaere *et al.* used TMP as a plasma (TMP*), which resolved its low reactivity and simultaneously promoted growth of the material. The films are fast growing (0.37 nm/cycle for aluminum phosphate) and have a higher P content (P/M of 1.8; O/P of 3.4), consistent with a pyrophosphate.[16] The main drawback of TMP* is the limitation in viable temperature. This occurs due to the polymerization of the TMP precursor as a plasma at lower temperatures which results in chemical vapor deposition (CVD)-like growth. To prevent this non-ALD behavior, the temperature window of these processes is limited to a narrow region around 300 °C.

One interesting alternative to TMP and TMP* could be tris (dimethylamino)phosphine (TDMAP). In a recent review on the deposition of metal phosphate processes using ALD,[6] TDMAP emerged as a viable candidate to deliver films that have a P/M ratio of 0.67–0.83 and a limited O/P ratio of approximately 3, indicative of P—O—P linkages in the material. However, these promising data originated from only one paper that performed a full characterization of a TDMAP-based magnesium phosphate process.[5] The process consisted of four different subsequent precursor steps starting with TDMAP, which was subsequently activated by an oxygen plasma (O_2*). This active surface was suited for reaction with the bis(ethylcyclopentadienyl)magnesium [$Mg(EtCp)_2$] precursor, which in turn was exposed to H_2O to complete the process. By repeating this sequence, a material was grown with a growth per cycle of approximately 0.115 nm/cycle at a temperature of 300 °C. Because this is the only report of a phosphate ALD process using TDMAP and of an ALD-based magnesium phosphate, it is not obvious to disentangle the role of the TDMAP precursor in achieving a polyphosphate matrix.

There have been two other reports using TDMAP. The first report was by Shibata,[17] who created an ALD lithium phosphorous oxynitride film with a P/M ratio of 0.33–0.5 and P/O of 3–4.5 (based on XPS results). The second report was written by Strempel *et al.*[18] They deposited a TDMAP-based aluminum phosphate, but this was done within the scope of doping a vanadia material with phosphorous, and they did not publish growth characteristics on their TDMAP-based aluminum phosphate process. Besides metal phosphate depositions, there was one additional reported use of TDMAP for the deposition of gallium phosphide by Graugnard *et al.*[19] in 2006.

To further introduce TDMAP as an alternative P-precursor for metal phosphate ALD processes, a process for the deposition of aluminum phosphate using TDMAP is developed and characterized in this work, based on the four-step ALD process of Su *et al.*[5] Here, repeated TDMAP, oxygen plasma (O_2*), trimethyl aluminum (TMA), and again oxygen plasma (O_2*) exposures are used to grow amorphous aluminum phosphate thin films. Aluminum is selected as the metal of choice because TMA is one of the most documented precursors in ALD literature. Moreover, the growth characteristics and film properties can be compared with previous reports on aluminum phosphate materials grown with TMP[14] and TMP plasma,[16] providing more information on the role of the phosphorous precursor. A supercycle strategy is also discussed in this work, which enables tuning the P content. Finally, the impact of the P content on the functionality of the deposited layers as a battery coating material is presented.

II. METHODS

A. ALD system

The system used for depositing the ALD materials was a home-built pump-type reactor. This system is equipped with viewports for ellipsometry and a plasma column. A more in depth description can be found in earlier work.[20] During depositions, the walls of the reactor were heated to 120 °C, and a base pressure of 5×10^{-6} mbar in between the different ALD steps was maintained. Depositions were performed using a substrate temperature ranging from 125 to 350 °C. If not mentioned, the substrate temperature was 300 °C. The substrate that was heated is always double polished silicon with a silicon-oxide layer at the surface due to continued exposure to the ambient air unless noted otherwise. An overview of all the precursors and the conditions applied during the ALD process is listed in Table I. Tris(dimethylamino)phosphine

TABLE I. Overview of the precursors and conditions used for the ALD depositions. RT = room temperature, which was (21±2) °C. Oxygen plasma was applied using a power of 200 W. When multiple O_2-plasma pulses were used in one process, the same conditions were used.

Parameter	TDMAP	O_2-plasma	TMA	H_2O
Pressure (mbar)	7.0×10^{-3}	1×10^{-2}	6.0×10^{-3}	8.0×10^{-3}
Exposure time (s)	15	10	15	15
Evacuation time (s)	35	45	40	45
Precursor heating (°C)	28	RT	RT	RT
Line heating (°C)	40	RT	70	70

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TABLE II. Composition of the as-deposited films analyzed using ERD. The deposition temperature was 300 °C.

Sample (ERD)	P(at. %)	O(at. %)	Al(at. %)	C(at. %)	H(at. %)	N (at. %)
TDMAP–O_2*–TMA–O_2*	11	64	23.4	0.1	1.5	0.0
TDMAP–O_2*–TMA–H_2O	12.1	62.7	23.1	0.1	1.9	0.0

(TDMAP) was supplied by Sigma-Aldrich, purified O_2-gas was delivered by Air Liquide, and TMA was packaged in a Strem supplied container and filled by Volatec.

B. Material characterization

1. Thickness measurements

In situ thickness measurements were conducted using a Filmsense Ellipsometer (SE). This ellipsometer measures the polarization at four different wavelengths (450, 550, 595, and 660 nm), and an optical model is fitted to the resulting data to extract the film thickness. The chosen model consisted of three layers: a heated substrate layer, a native oxide layer (ranging from 2 to 4 nm), and finally the ALD material that was described using a Cauchy model. The film thickness as well as the Cauchy parameters were fitted. To verify the SE results, complementary thickness measurements were conducted with x-ray reflectivity (XRR) (Bruker, Model D8) using Cu-Kα radiation. The XRR patterns were again fitted by considering a three-layer system. The obtained thickness values were used to verify the SE model.

2. Composition analysis

To obtain the composition of the materials, a combination of Electron Recoil Detection (ERD) and x-ray photoelectron spectroscopy (XPS) was used. ERD data in Table II were measured and analyzed at imec in Belgium using a 10 MeV $^{79}Br^{5+}$ ion beam. ERD measurements (Tables III and IV) were performed and analyzed at the Helmholtz-Zentrum Dresden Rossendorf using a 25 MeV $^{35}Cl^{5+}$ ion beam. The angle between the sample normal and the incoming beam was set to 75° with a scattering angle of 40°. The analyzed area was 2× 2 mm^2, and the recoiling sample atoms were detected using a time-of-flight energy telescope. The resulting mass resolving power is $M/\Delta M = 30$ if $M < 30$ u. Analysis of the data was performed using the program Potku[21] in slab analysis mode.

XPS was performed on a Thermofisher Scientific™Theta Probe instrument (10^{-10} mbar base pressure) using monochromatic Al Kα (1486.6 eV) radiation generated at 15 kV, focused into a 300 μm spot onto the sample surface and under an angle of 30° with respect to the normal on the sample surface. The input lens of the concentric hemispherical analyzer is at an angle of 50° with respect to the surface normal and accepts electrons over a wide angular range (20°–80° with respect to the surface normal). Survey scans were acquired at 200 eV pass energy with a 0.5 eV step size, and detail scans were acquired at 50 eV pass energy with a step size of 0.1 eV. Depth profiling was performed by rastering a (3 kV, 2 μA) Ar^+ ion beam, generated by the EX05 ion gun, under an angle of 45° to the surface normal and in an area of 2× 2 mm^2. The P 2s, Al 2p, and O 1s peak area were analyzed using the CASAXPS software package v2.3.26.[22] using the most suited background which was linear or Tougaard. Within the paper, the P atomic fraction is determined by dividing the peak area of P 2s by the summed areas of the P 2s, AL 2p, and O 1s signals, taking into account the relative sensitivity factors of the photo-emission peaks. The binding energy scale was referenced by setting the C 1s peak position to 284.6 eV. The materials do not inherently contain a significant amount of C; thus, the surface C that is present was formed by adventitious carbon.

3. Other characterization

Fourier Transform Infrared spectroscopy (FTIR) data were acquired (*ex situ*) with a Bruker Vertex V70 instrument, with a KBr beam splitter. Spectra were collected in transmission mode, in the range of 4000–400 cm^{-1}, coupled with a room temperature deuterated L-alanine doped triglycine sulfate (DLaTGS) detector. 400 scans with a resolution of 4 cm^{-1} were collected for each measurement. The measurement chamber was pumped down to ca. 10^{-2} mbar using a scroll pump prior to the collection of spectra. Si wafers polished on both sides were used to maximize the signal, and a blank Si substrate, taken from the same area of the Si wafer as the measured sample was used to obtain and subtract the

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TABLE III. Overview of the ERD results for the materials grown with different precursor combinations. Both the T_P–O and T_P–T_{Al} processes were not able to grow the required thickness (±80 nm) to properly conduct the ERD measurements and hence no data are reported.

Sample (ERD)	N (at. %)	P (at. %)	O (at. %)	Al (at. %)	C (at. %)	H (at. %)
TDMAP–O_2*–TMA–O_2*	0.36	11.6	63.6	21.1	0.11	3.02
TDMAP–O_2*–TMA	0	0.73	57.2	33.1	0.38	8.52
TDMAP–TMA–O_2*	0.32	11	64.1	22.2	0	2.37
TDMAP–O_2*	…	…	…	…	…	…
TDMAP–TMA	…	…	…	…	…	…

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TABLE IV. Overview of the ERD results for the materials grown with different supercycles.

Sample (ERD)	N (at. %)	P (at. %)	O (at. %)	Al (at. %)	C (at. %)	H(at. %)
AlO–PO (1-1)	0.36	11.6	63.6	21.1	0.11	3.02
AlO–PO (1-3)	0.41	16	65	12.5	0	6.10
AlO–PO (1-6)	0.47	18.4	66	11.1	0	3.97

background. The data were collected and processed using the Bruker OPUS software package.

X-ray diffraction (XRD) *in situ* and *ex situ* measurements were performed on a Bruker D8 diffractometer with a Cu-Kα x-ray source and a linear detector. *In situ* measurements can be conducted during heating under different atmospheres (H_2, air, and He).

Electrochemical characterization experiments took place in Ar atmosphere with both H_2O and O_2 levels kept below 1 ppm. The used potentiostat was home-built according to the specifications that were previously published.[23] The used cell consisted of a PTFE body that was filled with electrolyte [1M $LiClO_4$ in propylene carbonate (Sigma-Aldrich)] and connected to the working electrode (Si/20 nm SiO_2/40 nm PVD TiN/40 nm TiO_2), which had the materials of interest deposited on top. The exposed surface area was 1.05 cm^2. Li (Sigma-Aldrich) was used as both the counter and reference electrodes.

III. RESULTS

A. ALD process

1. Growth of aluminum phosphate using TDMAP

In this work, we adapted the four-step magnesium phosphate ALD process that was reported by Su *et al.*[5] and consisted of TDMAP–O_2*–Mg(EtCp)2–H_2O. The $Mg(EtCp)_2$ step was replaced with TMA to obtain aluminum phosphate as the final product. An overview of the precursor conditions that were used can be found in Table I.

ERD characterization of films grown with a TDMAP–O_2*–TMA–H_2O process confirmed the deposition of an aluminum phosphate material with ca. 12 at. % of phosphorous (Table II). Using *in situ* ellipsometry, it was observed that the growth of the material showed some deviations from linear growth as is shown in Fig. S1 in the supplementary material. By replacing the H_2O with an O_2* step, a more consistent growth of the material was observed [Fig. 1(a)], while this change did not have a significant impact on the composition of the material (only 1% decrease in P content and 1.3% increase in O content according to the ERD data in Table II).

In Fig. 1(b), a zoomed-in view of the growth curve is shown, detailing the thickness after each precursor pulse. From this, it can be observed that both TDMAP and TMA add to the growth of material while the O_2* pulses lead to a decrease in thickness. The reduction is due to the removal of the precursor ligands, creating reaction sites for the TMA and TDMAP molecules delivered in the subsequent ALD step.

Using the SE data from ALD cycles 75–100 (to prevent influences from substrate effects), the average added thickness was calculated to be 0.20 and 0.24 nm/pulse for TDMAP and TMA, respectively. The first O_2* pulse removed on average 0.16 nm/pulse and the second 0.17 nm/pulse. This yields a net growth of 0.11 nm/cycle; accounting for further experiments, a GPC value of 0.1104 $\pm$ 0.0090 nm/cycle was obtained (0.0090 nm being the standard deviation determined from 13 different experiments).

To study the ALD-like nature of this TDMAP–O_2*–TMA–O_2* process (i.e., T_P–O–T_{Al}–O), *in situ* ellipsometry was performed while the exposure time of each precursor was varied to obtain saturation curves. To verify the SE data, XRR measurements were performed and these were found to be in good agreement for a

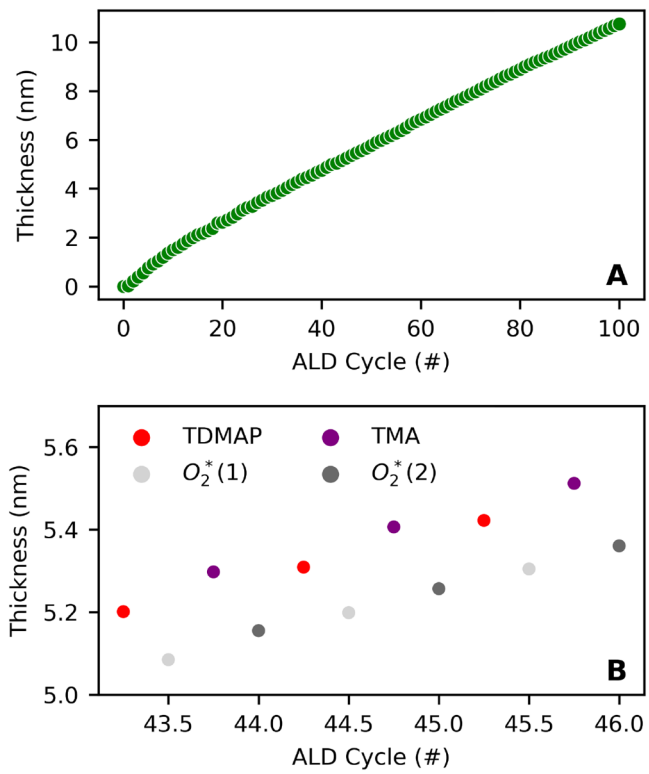

FIG. 1. (a) *In situ* ellipsometry data of the growth of a TDMAP–O_2*–TMA–O_2* process. The deposition temperature was 300 °C. (b) A close-up of the thickness measured after each precursor exposure for three subsequent cycles.

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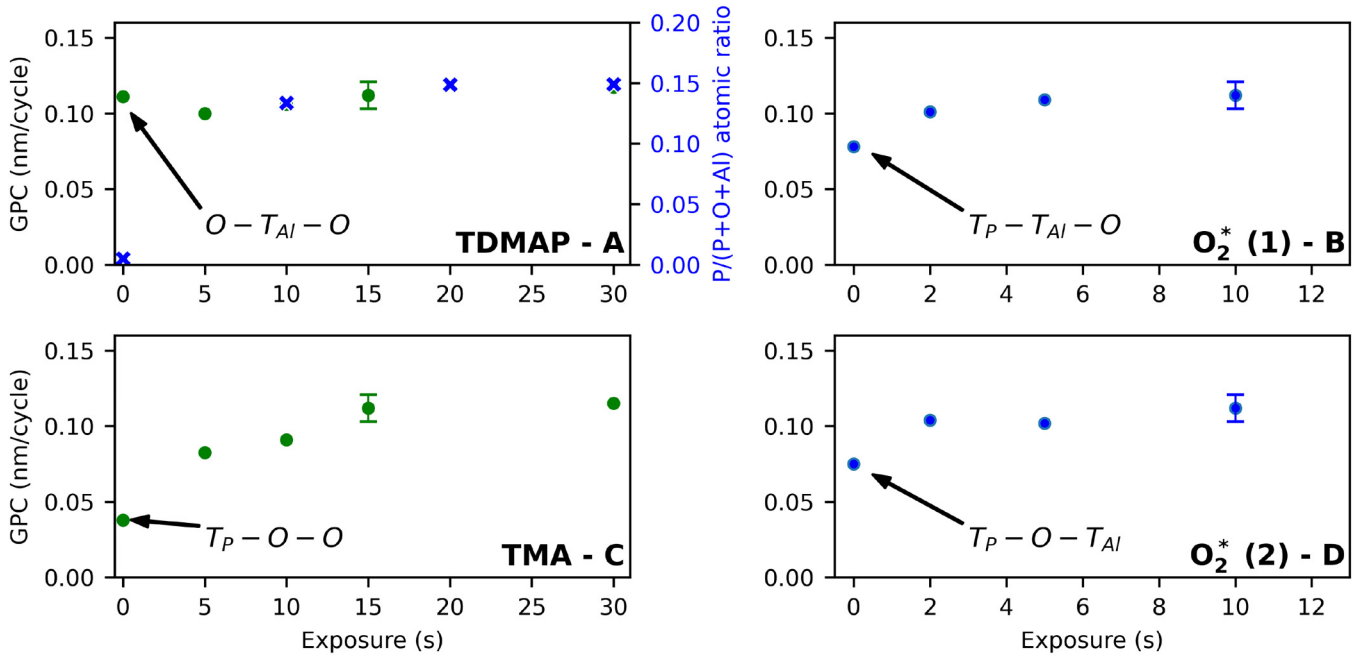

FIG. 2. Overview of the different saturation curves of the four-step process T_P–O–T_{Al}–O at 300 °C. The error bar is displayed for the conditions that were set while saturation was tested. It was calculated by taking the standard deviation of 13 experiments that were performed throughout the experimental phase of the work. (a) The secondary y-axis of the TDMAP saturation curve indicates the P atomic fraction, obtained from XPS. (b) Saturation behavior of the first O_2-plasma pulse. (c) Saturation behavior of TMA. (d) Saturation behavior of the second O_2-plasma pulse.

refractive index of 1.50 ± 0.05 (at 633 nm), which is similar to previous aluminum phosphate reports[14,16]).

In these saturation experiments, the exposure time of one precursor was varied while those of the three other precursors were fixed. The fixed exposure times were 15, 10, 15, and 10 s for TDMAP, first O_2* pulse, TMA, and second O_2* pulse, respectively. The results of these experiments are shown in Fig. 2. All of the four different precursor steps reach saturation. For TDMAP, the data show no real difference in the growth rate with or without TDMAP (both 0 and 30 s have a GPC of 0.11 nm/cycle). This is due to the similar GPC of the T_P–O–T_{Al}–O process in saturated conditions and the TMA–O_2* process for aluminum oxide.[24] To verify that not only the growth but also the incorporation of phosphorous is saturated, the P atomic fraction was derived from XPS, saturation in P content seems to be reached already after 10 s of TDMAP exposure. The other curves show an expected saturation after 2, 15, and 2 s for the first O_2*, TMA, and the second O_2* pulses, respectively. To ensure saturated conditions, slightly higher exposure times (see Table I) were used for the depositions in this work. When the first O_2* pulse, TMA, or the second O_2* pulse is eliminated from the deposition sequence (exposure = 0 s in the respective saturation plots), a lower growth is observed. These three-step processes are discussed in more detail in Sec. III A 2.

An additional experiment where TDMAP was repeatedly pulsed over a bare Si and an aluminum phosphate surface was performed to ensure that the TDMAP precursor does not grow nor decompose without the use of O_2* as a coreactant. The SE curve in Fig. S2 in the supplementary material confirms negligible growth when only TDMAP is pulsed.

There is a decrease in GPC with substrate temperature as can be seen in Fig. 3(a). The GPC decreased from 0.20 nm/cycle at 125 °C to 0.08 nm/cycle at 350 °C. According to the *in situ* SE measurements, this decrease in growth rate with increasing temperature can be attributed to both TDMAP and TMA contributes a lower growth per pulse (from an added thickness of 0.28–0.12 nm/cycle for TDMAP and from 0.30 to 0.19 nm/cycle for TMA when the temperature increased from 125 to 350 °C). To verify that there is no condensation of the TDMAP precursor at the lower temperature ranges, an additional saturation experiment was performed at 125 °C with an extended TDMAP exposure of 60 s. This resulted in a marginal increase of the GPC from 0.193 to 0.199 nm/cycle. The composition of materials deposited at selected temperatures was analyzed using XPS [see secondary y-axis of Fig. 3(a)] showing no clear temperature dependency of the P atomic fraction. Figure 3(b) shows the average thickness addition per precursor pulse, as obtained from ellipsometric data that were acquired after each precursor exposure. With increasing temperature, the average pulse effect decreases. Less growth is seen for both the TMA and TDMAP exposures, while also the effect of the O_2* steps decreases with increasing temperature. This is in line with the overall decrease in GPC seen in Fig. 3(a). This result reminds us of the TMA–H_2O and TMA–O_2* ALD processes, where this decrease in

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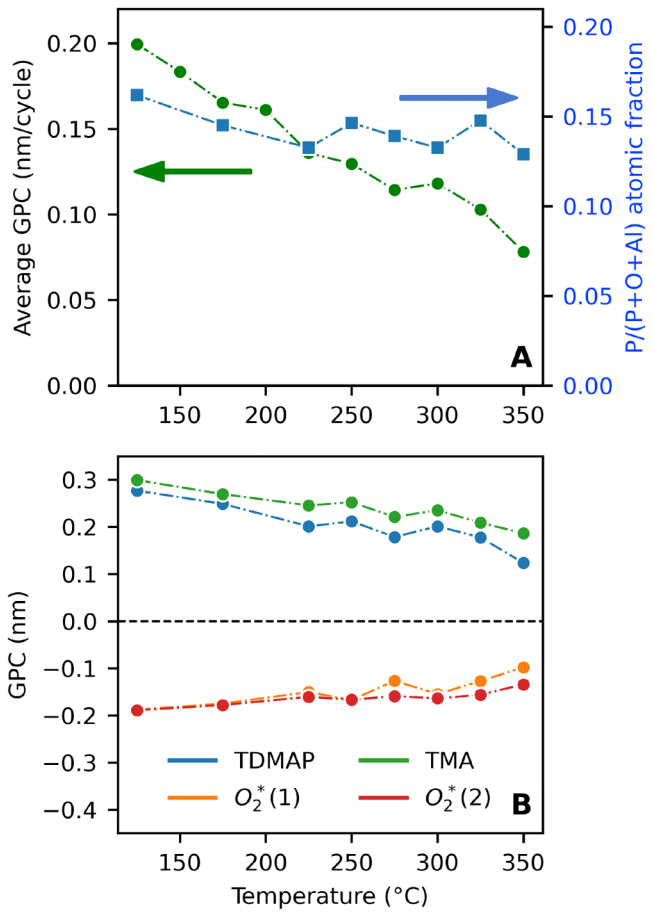

FIG. 3. (a) GPC against the deposition temperature. GPC values were calculated similarly to the saturation values, the values shown are the observed GPC in the last 25 cycles of each experiment. The secondary y-axis shows the P fraction measured using XPS. (b) Overview of the growth of each step of the process measured at different temperatures.

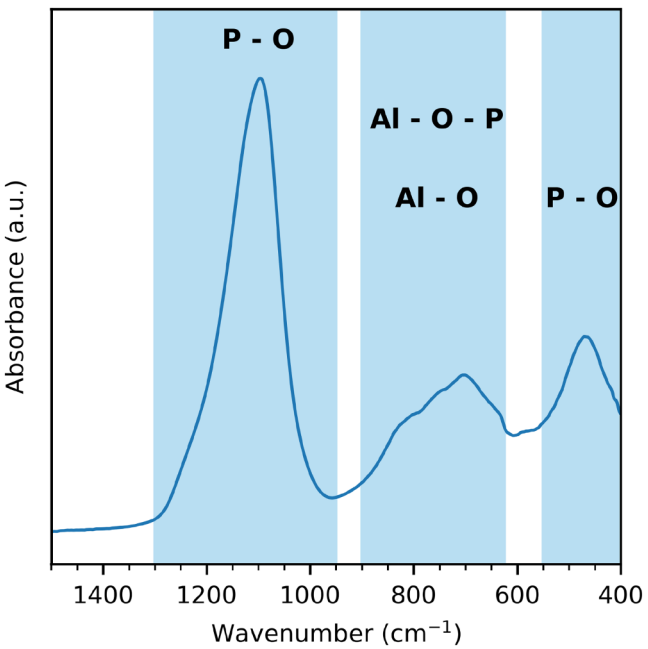

FIG. 4. 1500–400 cm^{-1} region of the FTIR spectra measured for the material deposited using the T_P–O–T_{Al}–O process. Indicated are the three regions that were associated with the presence of $AlPO_4$-tridymite structures (Ref. 30).

GPC is typically attributed to the desorption of reactive surface hydroxyl groups with increasing temperature.[25,26]

ERD analysis of the material deposited by the T_P–O–T_{Al}–O process at 300 °C yields a P/M ratio of 0.51 ± 0.04 and an O/P ratio of 5.65 ± 0.3 (based on two samples deposited and measured at different times, data in Tables II and III). This stoichiometry suggests that our material is a P-doped metal oxide rather than a fully fledged phosphate material (for $AlPO_4$ orthophosphate, P/M = 1 and O/P = 4). The nature of the P bonds was further investigated by FTIR. Figure 4 shows a zoomed in view of the 1500–400 cm^{-1} region. Interpretation of the exact peak position of inorganic materials is not straightforward due to numerous effects that can have an effect on the position and number of available bands.[27] We observe three broad bands in the spectrum. Two of these regions 1300–950 and 550–400 cm^{-1} are associated with P—O interactions, while the region from 625 to 900 cm^{-1} is attributed to Al—O and Al—O—P interactions. This claim is based on multiple sources.[27–29] One reference focusing on tridymite $AlPO_4$ observed more well-defined peaks with a smaller width at 1150–1100, 740–710, and 510–460 cm^{-1}. These were attributed to the asymmetric stretching vibration of P—O bonds in the $[[PO_4]^{3-}]$- tetrahedrons, the Al—O—P stretching vibration, and the P—O bending vibrations, respectively.[30]

An XPS-survey measurement was also performed and is shown in Fig. S3 in the supplementary material. This indicated that besides Al, P, and O (detailed spectra are available in Fig. S4 in the supplementary material), only C was observed on the surface of the film. This C was formed at the material's surface by adventitious carbon and was fully removed after sputtering as is clear from Fig. S5 in the supplementary material, which shows the C 1s peak before and after 100 s of sputtering in the XPS. The low C and N content in the films are also confirmed by the ERD results in Tables II and III, suggesting that the TDMAP ligands were removed. Note that one of the ERD measurements showed a 0.34% nitrogen content most likely due to sample air exposure. The uniformity of the composition throughout the film is also further illustrated by the ERD-depth profile in Fig. S6 in the supplementary material.

Both as-deposited and annealed thin films (in air, He, and H_2 up to 1000 °C) were amorphous, as concluded from the XRD plots in Fig. S7 in the supplementary material, which do not indicate the formation of any distinct crystal phases.

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2. Effect of coreactant

As was already observed in Sec. 1 (H_2O vs O_2* as coreactant), and as has been previously observed in the literature, the coreactants play an important role in the phosphate ALD process. For that reason, different variations of the T_P–O–T_{Al}–O process were attempted and characterized. The tested variations were T_P–T_{Al}–O (TDMAP–TMA–O_2*), T_P–O–T_{Al} (TDMAP–O_2*–TMA), T_P–O (TDMAP and O_2*) and T_P–T_{Al} (TDMAP–TMA). All are depicted, with their *in situ* growth measured using SE, along with the previously discussed four-step TDMAP–O_2*–TMA–O_2* process in Fig. 5. By comparing the compositions in Table III, interesting differences can be observed between these processes.

T_P–O–T_{Al}–O has the highest GPC and a P content of 11%. This is very similar to the composition of the T_P–T_{Al}–O process. The additional O_2* pulse, thus, stimulates the growth of the material while it does not have a significant effect on the composition of the material. TMA must have enough inherent reactivity to react with the TDMAP terminated surface. The T_P–O–T_{Al} process has a similar growth rate as the T_P–T_{Al}–O but the composition indicates that this process produces merely aluminum oxide rather than a phosphate material. This shows that while TMA is capable of reacting with the TDMAP terminated surface, the opposite does not seem to be feasible. In other words, TDMAP was introduced but left the system without the occurrence of chemisorption. Only after the subsequent introduction of O_2* does further growth occur. The low reactivity of a TDMAP–TMA cycle is also illustrated by the data obtained for an experiment that was performed without any coreactant steps. There is no sign of growth after the initial thickness increase of 0.37 nm after one cycle. This initial growth could be attributed to a singular T_P–T_{Al} interaction (as was observed during the T_P–T_{Al}–O process) but this sub-1 nm layer was not investigated further. The final experiment of T_P–O does show some growth until a saturation is reached at a total thickness of approximately 2 nm. Studying the composition of this layer or its surface would require *in situ* FTIR or *in vacuo* XPS methods, because phosphorous oxide is known to be hygroscopic, dissolving in ambient moisture to phosphoric acid. Therefore, no further *ex situ* characterization was done on this material.

An attempt to grow the material thermally was performed by using H_2O twice as the coreactant. This process had a very limited growth of 0.02 nm/cycle and although XPS measurements indicated that 8% phosphorous was present further experiments were not conducted.

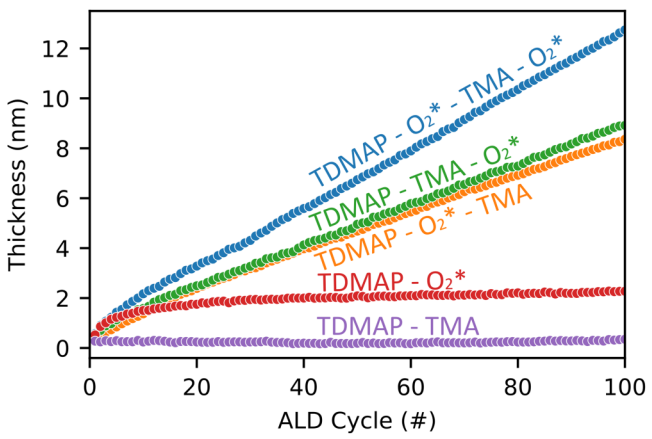

FIG. 5. *In situ* ellipsometry data for the different combinations of precursors that were tested at 300 °C. Depicted are the thicknesses obtained from the SE measurements conducted at the end of each full cycle.

3. Discussion: Comparison with other metal phosphate ALD processes

In this section, a comparison of our material with other reported aluminum phosphate materials is presented. Strempel *et al.*[18] previously deposited a similar aluminum phosphate using TDMAP. A substrate temperature of 150 °C was set and they reported a GPC of 0.131 nm/cycle, which is slightly lower than the 0.17 nm/cycle that was observed in this work for the same temperature. A reason for this difference is hard to determine since their report did not comment on the composition of the material and was deposited in a flow-type reactor, unlike the pump-type used in this work. The TMP–O_2*–TMA–O_2* process reported by Hornsveld *et al.* has a more comparable GPC at 150 °C of 0.17 nm/cycle but they saw little differences in GPC over the tested temperature range (25–300 °C).[14] The TMP* based process, on the other hand, has an inherently very high GPC of approximately 0.4 nm/cycle.[16] The obtained refractive index of 1.5 is also in agreement with the previously reported values.[14,16,31]

Regarding the composition of the deposited materials, an overview was made comparing P/M and O/P values of all metal phosphate materials that were deposited at 300 °C (using a metalorganic precursor—coreactant (optional)—phosphorous precursor—coreactant (optional) ALD sequence) (Fig. 6). Within both the Al-based phosphates as the phosphate class in general, the obtained P/M ratio of 0.51 is low while the O/P value of 5.65 is very high, meaning that the as-deposited material is quite far from an orthophosphate (P/M of 1 and O/P of 4). This is a stark difference with the other TDMAP-based phosphate process,[5] namely, for magnesium which had a P/M ratio of 0.67 and an O/P ratio of 3.16. Due to the lack of other reports on magnesium phosphate, it is difficult to pinpoint the origin of this difference. It is known that the choice of metal precursor in itself can dictate the growth behavior of ALD layers,[3] which is one hypothesis for the observed difference.

The structure of the as-deposited coating of these materials is amorphous. However, Dobbelaere was able to obtain a tridymite $AlPO_4$ structure by annealing the as-deposited material to 900 °C.[16] Hämäläinen *et al.* obtained the same crystalline phase after annealing, but used a different ALD process ($AlCl_3$–TMP).[31]

Based solely on our report and the comparison with the other phosphate materials, it would indicate that TDMAP incorporates a low amount of phosphorous in comparison with the alternative options: trimethylphosphate (TMP), TMP*; diethylphosphoramidate (DEPA) and DEPA*. P_2O_5 was not displayed because these depositions were reported using a very high substrate temperature of 500 °C.[32,33]

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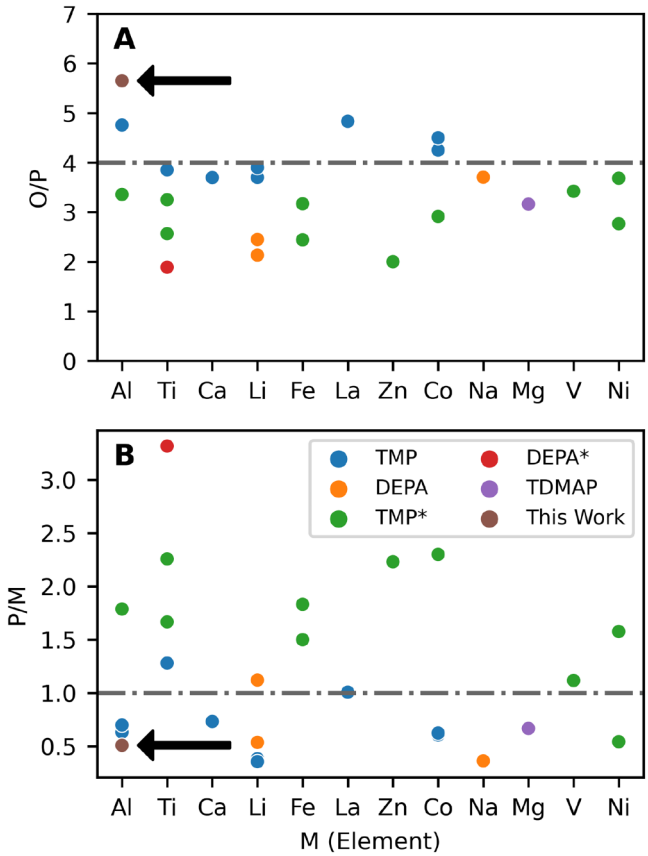

FIG. 6. (a) Overview of the O/P ratio of various metal phosphate materials. (b) Overview of the P/M ratio of various metal phosphate materials deposited using different phosphorous precursors. Our results are indicated with black arrows. Data for this table can be found in Table S1 in the supplementary material.

4. Tailoring of P content using the AlO–PO approach

To control the P content of the aluminum phosphate material, the same strategy that was previously used by multiple other reports was adapted.[14,32–34] The four-step process was split into two separate cycles, TDMAP–O_2* and TMA–O_2*. By changing the ratio of these individual cycles in a supercycle approach, a greater amount of phosphorous could be incorporated. This method also circumvented the saturation in growth that was previously observed for the T_P-O process (Fig. 5). Instead of saturating at a thickness of 2 nm, the introduction of the TMA–O_2* cycle seems to reset the surface enabling further growth of the TDMAP–O_2* cycles. This enables us to control the amount of phosphorous within the material in a similar way to the previously reported TMP-based process by Hornsveld *et al.*[14] An overview of the compositions of the deposited mixed materials is given in Table IV, while the *in situ* growth curves of these supercycle processes are given in Fig. 7. Clearly, the growth per supercycle and P content increase when the number of TDMAP–O_2* cycles in the supercycle increases from 1 to 3 and 6 (with the number of TMA–O_2* cycles fixed to 1).

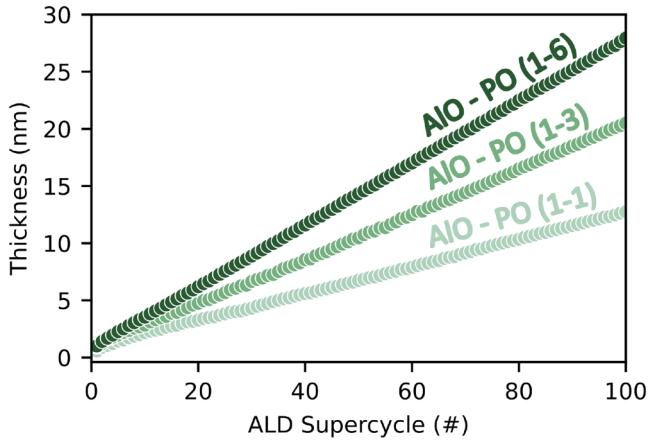

FIG. 7. *In situ* ellipsometry data of three processes with different AlO–PO ratios. As the contribution of PO increases, the overall growth of the material also increases.

Figure 8 shows the FTIR spectrum of a film grown with the 1-1 process (see also Fig. 4) together with the spectra of the 1-3 and 1-6 AlO–PO samples. As the number of PO cycles increases, the intensity in the regions associated with P—O interactions increases. The region associated with Al—O and Al—O—P bonds shows a diminished intensity, which can be attributed to the

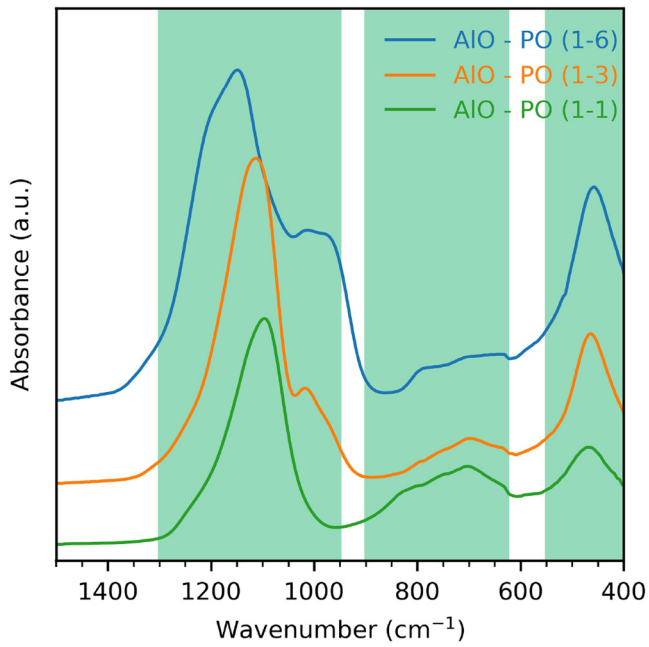

FIG. 8. Comparison between the respective FTIR spectra of the materials with AlO–PO ratios increasing from 1-1 to 1-6.

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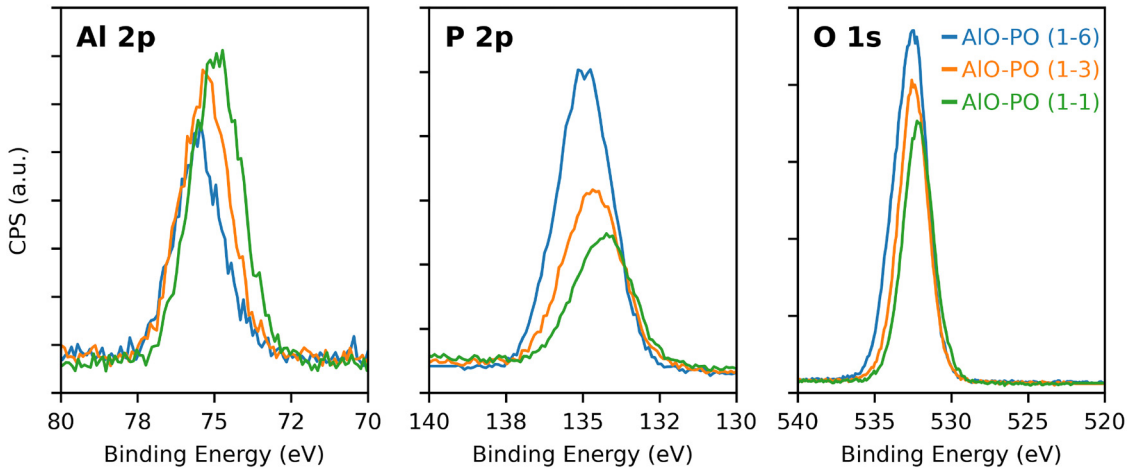

FIG. 9. XPS spectra for the materials grown with a different supercycle ratio measured at the surface of the sample. Binding energy calibration was performed on the basis of the C—C/C—H component of the C 1s peak.

relatively lower Al content of these films. The peaks at 1000 and 1200 cm^{-1} show a particularly sharp rise in intensity, which is attributed to the increasing prevalence of P—O—P bonds.[6,33,35] The higher P content in ERD and these FTIR results both indicate that the deposited material shifts from a P-doped oxide to an ortho- or polyphosphate material. In Fig. S8 in the supplementary material, an overview of the full spectra from 4000 to 400 cm^{-1} is given.

The same samples were also analyzed using XPS. Overlays of the individual Al 2p, P 2p, and O 1s peaks in Fig. 9 also indicate that there are some changes occurring in the structure of the material.

The Al 2p peak mildly shifts from 74.9 to 75.5 eV, where a shift toward higher binding energies indicates a higher P content.[36] The P 2p peak is an envelope peak comprising of the $2p_{3/2}$ and $2p_{1/2}$ components, which are close in binding energy (0.5–1 eV). The shift of this envelope peak toward higher binding energies, as the overall amount of phosphorous increases, has already been reported.[37] The higher binding energy was attributed to an increase in P═O bonds.

The O 1s peak position does not really shift, instead there is an increased intensity of a shoulder peak linked to the presence of bridging oxygen (P—O—P) at a slightly higher binding energy,[38–40] which causes a general broadening of the peak. This also indicates a shift from a P-doped oxide to a phosphate material.

Overall, a decrease in Al 2p peak intensity and an increase in P 2p and O 1s are observed as the number of PO cycles increases. This is also in agreement with a more P-rich environment (less Al—O and more P—O bonds).

On the XRD plots of the 1-3 and 1-6 samples (in Fig. S9 in the supplementary material) three distinct peaks associated with the $AlPO_4$-tridymite phase start forming at 870 °C in a helium atmosphere. The result of the 1-3 and 1-6 crystal phase is, thus, more reminiscent of the material that was reported by Dobbelaere *et al.*[16]

B. TDMAP based aluminum phosphate as battery electrode coating material

Within the field of battery coatings, there have been multiple reports on the use of aluminum phosphates as coatings.[4,9,10] Some materials that have been deposited before using ALD on battery electrodes include AlPO(N),[4] Li_3PO_4—TiO_2,[41] and $FePO_4$.[42] All of these were deposited using a TMP, TMP*, or dimethylphosphoramidate-plasma (DEPA*). To our knowledge, the effect of varying the P content in the coatings was never systematically addressed. The dual cycle method that was just discussed offers the possibility of tailoring the P content within a certain range.

The Li-ion battery electrode material that was coated is a 40 nm TiO_2 thin film on a sputtered TiN substrate. TiO_2 is a model system that is used to evaluate the Li-ion transport through the deposited AlPO coating into the TiO_2 bulk during the lithiation step and out of the TiO_2 in the delithiation step. The previously mentioned supercycle approach was used to deposit three AlPO coatings with a thickness of 2 nm. The films were characterized by performing a CV-sweep and checking the lithiation capacity of the different films.

Figure 10 shows the electrochemical data. The lithiation step in the anodic sweep has a peak at 1.65 V, indicating the reduction of Ti from Ti^{+IV} to Ti^{+III} as Li is migrating into the TiO_2 bulk to form $LiTiO_2$. The delithiation in the cathodic sweep has a main peak at 2.25 V denoting the migration of the Li-ion out of the electrode oxidizing Ti^{+III} back to Ti^{+IV}. There is also a minor peak visible around 1.65 V in the cathodic sweep, and this peak is associated with the transfer of Li ions in oxygen vacancies in the TiO_2 structure.[43]

The lithiation step of the 1-1 material indicates that Li-ion transport was severely limited by application of the AlPO coating. By adding more of the TDMAP–O_2* steps, the phosphorous content is increased and this has a positive effect on the coating

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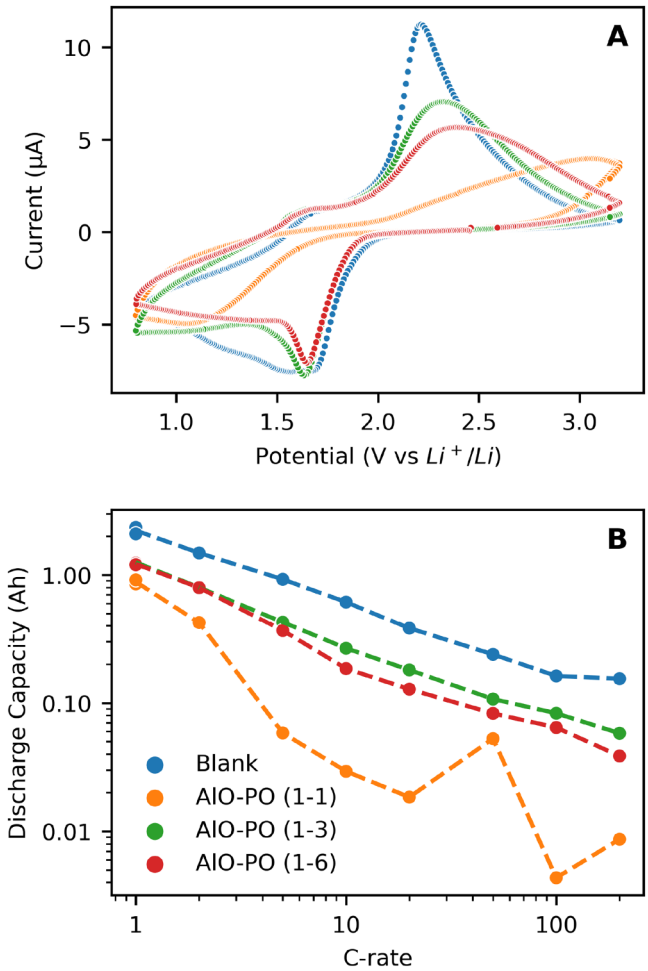

FIG. 10. (a) Cyclic voltammograms of TiO_2 substrates coated with a thin layer (approximately 2 nm) of a TDMAP-based aluminum phosphate film with varying AlO:PO ratios. The scan rate used was 1 mV/s, and the scans were conducted from 0.8 to 3.5 V (vs Li^+/Li) (b) Discharge capacities for different C-rates (1 C indicates charging speed equal to the batteries capacity, 2 C indicates twice as fast, etc.). The current at 1 C = 2.4 μA.

material. The best result was obtained for the 1-3 ratio rather than the 1-6, signifying that there is a composition that has optimal performance. These results are also corroborated by the discharge capacities for different C-rates in Fig. 10(b), with the 1-3 sample showing the best performance.

This work clearly signifies that the P content plays a vital role in developing the most optimal phosphate coatings. Future works could take this into account when developing metal phosphate coatings for Li-ion battery electrodes.

IV. CONCLUSION

In this work, TDMAP has been established as a viable phosphorous precursor. This was done by performing a full characterization of the four-step ALD process consisting of TDMAP–O_2*–TMA–O_2*. The material deposited at 300 °C could be grown in a self-limiting manner with a GPC of approximately 0.11 ± 0.02 nm/cycle. It has a P/M content of 0.51 ± 0.04 and an O/P ratio of 5.65 ± 0.3. No nitrogen or carbon incorporation was detected (in the bulk of the material) during the ERD analysis of the samples indicating that these are completely removed from the film.

The aluminum phosphate material was deposited in a self-limiting manner at different temperatures ranging from 125 to 350 °C. The GPC varied between 0.20 nm/cycle at 125 °C and decreased to 0.08 nm/cycle at 350 °C. The temperature used for most of the analysis was 300 °C where an average GPC of 0.1104 ± 0.0090 nm/cycle was calculated. The as-deposited material was amorphous and did not crystallize during annealing in different atmospheres (air, He, H_2) up to the instrumental limit of 1000 °C when measured with XRD. However, analysis of the XPS and FTIR data of the as-deposited material does suggest the presence of tridymite $AlPO_4$ clusters in the material.

By testing different combinations of coreactants, information on the interactions between TDMAP and TMA was obtained, showing that TMA is able to interact with a TDMAP terminated surface while the reverse reaction does not occur.

The four-step process can be considered a supercycle process comprising one ALD cycle consisting of TDMAP– O_2* and oe cycle of TMA–O_2*. By changing the ratio of the ALD cycles within the supercycle, substantial changes in composition were obtained. When combining six TDMAP–O_2* cycles with one TMA–O_2* cycle, the P/M ratio could be increased to 1.66 and the P/O ratio could be reduced to 3.59, both of which indicate the formation of more phosphorous containing phosphate structures such as pyro- and metaphosphates. These materials were also able to crystallize to $AlPO_4$-tridymite at 870 °C as was confirmed by XRD.

Comparing these materials as thin film coatings for a battery electrode showed a clear indication that the P content of phosphate materials has a significant impact on the performance of the coating.

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SUPPLEMENTARY MATERIAL

The following content is provided in the supplementary material: (S1) A side-by-side comparison, using *in situ* ellipsometry, of the growth of 100 ALD cycles of TDMAP–O_2*–TMA–O_2* and TDMAP–O_2*–TMA–H_2O. (S2) *In situ* ellipsometry data to indicate that TDMAP does not decompose on different surfaces of Si or aluminum-phosphate. (S3) XPS-survey spectrum of the TDMAP–O_2*–TMA– O_2* process at 300 °C. (S4) High resolution XPS data of the Al 2p, P 2p, and O 1s peaks for the TDMAP–O_2*–TMA–O_2* process at 300 °C. (S5) High resolution XPS data for the C 1s peak before and after sputtering for the TDMAP–O_2*–TMA–O_2* process. (S6) ERD data highlighting the conformatlity of the TDMAP–O_2*–TMA–O_2* process throughout an 80 nm sample. (S7) *In situ* XRD data of the TDMAP–O_2*–TMA–O_2* process annealed in different atmospheres. (S8) Full range (4000–400 cm^{-1}) FTIR spectra for all measured samples (AlO–PO). (S9) *In situ* XRD of the 1-3 AlO–PO sample annealed in a He atmosphere. (Table S1) Overview of all the data used for the compilation of Fig. 6.

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AUTHOR DECLARATIONS

Conflict of Interest

The authors have no conflicts to disclose.

Author Contributions

Ruben Blomme: Conceptualization (supporting); Data curation (lead); Formal analysis (lead); Investigation (lead); Methodology (lead); Validation (lead); Visualization (lead); Writing – original draft (lead); Writing – review & editing (lead). **Aditya Chalishazar:** Data curation (supporting); Investigation (supporting); Visualization (supporting); Writing – original draft (supporting); Writing – review & editing (supporting). **Lowie Henderick:** Data curation (supporting); Writing – original draft (supporting); Writing – review & editing (supporting). **Frans Munnik:** Data curation (supporting); Investigation (supporting); Resources (supporting); Writing – review & editing (supporting). **Johan Meersschaut:** Data curation (supporting); Investigation (supporting); Resources (supporting); Writing – review & editing (supporting). **Matthias M. Minjauw:** Data curation (supporting); Investigation (supporting); Resources (supporting); Writing – review & editing (supporting). **Christophe Detavernier:** Funding acquisition (supporting); Project administration (supporting); Resources (supporting); Supervision (supporting); Writing – review & editing (supporting). **Jolien Dendooven:** Conceptualization (lead); Data curation (supporting); Formal analysis (supporting); Funding acquisition (lead); Investigation (supporting); Methodology (supporting); Project administration (lead); Resources (lead); Supervision (lead); Validation (supporting); Visualization (supporting); Writing – original draft (supporting); Writing – review & editing (supporting).

DATA AVAILABILITY

The data that support the findings of this study are available from the corresponding author upon reasonable request.

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